

Problem

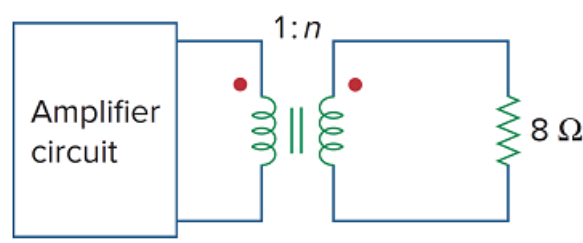
A transformer is used to match an amplifier with an $8\text{-}\Omega$ load as shown in Fig. 13.119. The Thevenin equivalent of the amplifier is: $V_{Th} = 10\text{ V}$, $Z_{Th} = 128\text{ }\Omega$.

(a) Find the required turns ratio for maximum energy power transfer.

(b) Determine the primary and secondary currents.

(c) Calculate the primary and secondary voltages.

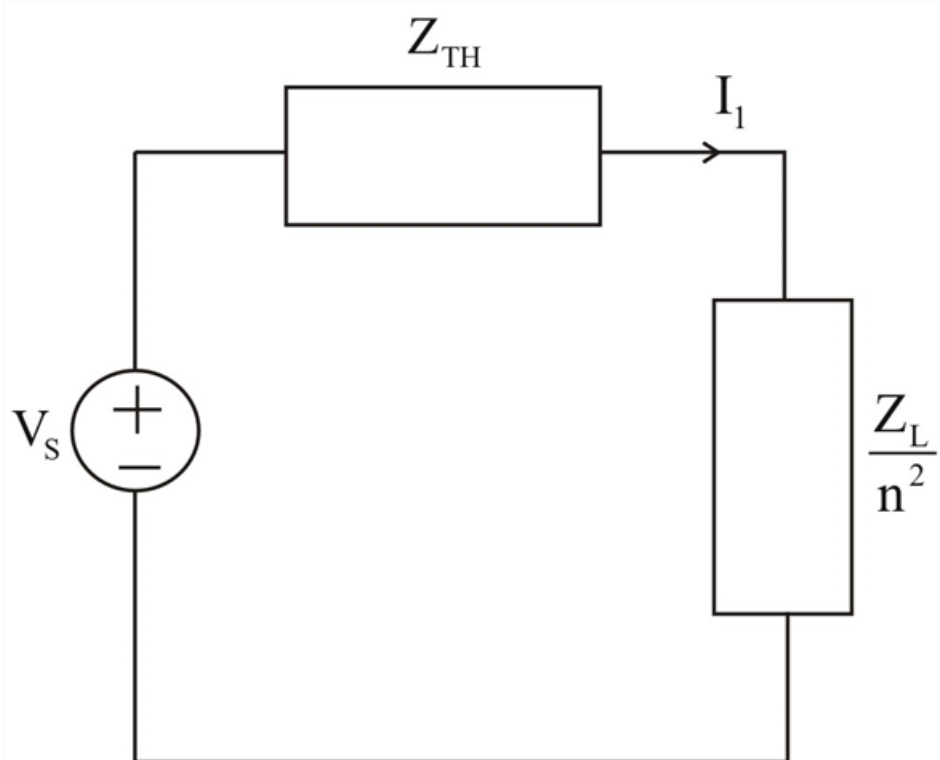
Figure 13.119



Step-by-step solution

Step 1 of 4

(A)



Step 2 of 4

For maximum power transfer,

$$Z_{TH} = \frac{Z_L}{n^2}$$

Or,

$$n^2 = \frac{Z_L}{Z_{TH}} = \frac{8}{128}$$

$$n = 0.25 \text{ .}$$

Step 3 of 4

$$\begin{aligned} \text{(B) } I_1 &= \frac{V_{TH}}{\left(Z_{TH} + \frac{Z_L}{n^2}\right)} = \frac{10}{(128+128)} \\ &= 39.06 \text{ } m A \text{ .} \end{aligned}$$

Step 4 of 4

$$\text{(C) } V_2 = I_2 Z_L = 156.24 \times 8 mV = 1.25 \text{ } V$$

But,

$$V_2 = nV_1$$

Therefore,

$$V_1 = \frac{V_2}{n} = 4(1.25) = 5V \text{ .}$$